



15 lines (12 loc) · 1.21 KB

Preview

Code

Blame

Raw



Real-Time ECG Monitoring System

Final Year Project Report

1. Abstract

This project describes the design and implementation of a low-cost, portable, Real-Time ECG Monitoring System. By utilizing the AD8232 ECG sensor module and the ESP8266 NodeMCU micro-controller, cardiac electrical activity is captured and transmitted wirelessly. The system provides immediate visualization of ECG waveforms for remote health tracking.

2. Hardware Architecture

The system hardware consists of two main components:

- **AD8232 Sensor Module:** An integrated signal conditioning block for ECG and other biopotential measurement applications designed to extract, amplify, and filter small biopotential signals in the presence of noisy conditions.
- **ESP8266 NodeMCU (ESP-12E):** A Wi-Fi enabled micro-controller responsible for analog-to-digital conversion (ADC) of the sensor data and handling wireless data protocols.

3. Software & Data Flow

1. The AD8232 outputs an analog signal representing the heart's electrical activity.
2. The ESP8266 reads this signal via its analog pin (A0) at a sampled frequency.

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3. The data is processed locally and streamed over the Arduino IDE Serial Plotter for real-time waveform visualization.